

Livestock Disease Detection from Images Using Convolutional Neural Networks and Transfer Learning Techniques

Phase 2 – Proposal and Code Implementation

Submitted By

Bimal Tenson

M.Sc. Data Science

Course: Machine Learning

Project Phase: Phase 2 – Proposal and Code Implementation

Submission Date: June 2026

University of Europe for Applied Sciences

Master of Science in Data Science

1. Project Title
2. Background and Motivation
3. Problem Statement
4. Dataset Description
5. Research Questions
6. Expected Results
7. Proposed Methodology
8. CNN Model Design
9. Transfer Learning Model
10. Evaluation Metrics
11. Expected Figures and Tables
12. References

1. Project Title

Livestock Disease Detection from Images Using Convolutional Neural Networks and Transfer Learning Techniques

2. Background and Motivation

Livestock plays a vital role in the agricultural sector and contributes significantly to food production and the economy. Diseases affecting cattle can lead to reduced productivity, financial losses for farmers, and potential risks to food security. Early detection of livestock diseases is therefore essential for effective treatment and disease management.

Traditionally, disease diagnosis relies on veterinary expertise and manual observation, which can be time-consuming and may not always be available in remote areas. With recent advancements in Artificial Intelligence (AI) and Deep Learning, image-based disease detection has emerged as a promising solution. Convolutional Neural Networks (CNNs) have shown remarkable performance in image classification tasks and can assist in identifying diseases from livestock images with high accuracy.

This project aims to develop an automated livestock disease detection system using deep learning techniques. The study explores the effectiveness of CNN models in classifying cattle diseases from images and investigates the potential of transfer learning architectures for improving performance.

3. Problem Statement

Livestock diseases such as Lumpy Skin Disease and Foot-and-Mouth Disease can spread rapidly and cause significant economic losses. Early and accurate disease identification is often challenging due to limited veterinary resources and delays in diagnosis.

The primary problem addressed in this project is the development of an automated image-based classification system capable of accurately identifying cattle diseases from photographs. The proposed solution utilizes Convolutional Neural Networks to classify cattle images into different disease categories and assist in faster disease detection.

4. Dataset Description

The dataset used in this project consists of cattle images collected and organized into three categories representing different health conditions.

Dataset Statistics

- Total Images: 3,244
- Number of Classes: 3

Class Distribution

Class	Number of Images
Healthy	1,291
Lumpy Skin Disease	1,207
Foot-and-Mouth Disease	746
Total	3,244

Image Type

- RGB color images
- Cattle disease images
- Used for multi-class image classification

The dataset contains images representing healthy cattle and cattle affected by different diseases. These images are used to train and evaluate deep learning models for disease classification.

5. Research Questions

1. Can a Convolutional Neural Network accurately classify livestock diseases from image data?
2. How effectively can the CNN model distinguish between healthy cattle and diseased cattle?
3. Which disease category is detected with the highest classification accuracy?
4. How does a custom CNN model compare with a MobileNetV2 architecture in livestock disease detection?

5. What challenges and classification errors occur during disease prediction?

6. Expected Results

The expected outcome of this project is the development of an effective deep learning model capable of automatically classifying cattle diseases from images.

Expected results include:

- Accurate identification of healthy and diseased cattle.
- Improved disease detection through automated image analysis.
- Classification accuracy is greater than 70%.
- Generation of confusion matrices and classification reports for performance evaluation.
- Comparison of CNN and MobileNetV2 architectures.
- Demonstration of the usefulness of deep learning in livestock healthcare applications.

7. Proposed Methodology

The project follows a supervised deep learning approach for image classification.

The methodology consists of the following stages:

1. Dataset Collection and Preparation
 - Load cattle disease image dataset.
 - Organize images according to disease classes.
2. Data Preprocessing
 - Resize all images to 224×224 pixels.
 - Normalize pixel values.
 - Apply data augmentation techniques.
3. Dataset Splitting
 - Training Set: 80%
 - Validation Set: 20%
4. Model Development
 - Design and implement custom CNN architecture.
 - Train the model using the prepared dataset.
5. Model Evaluation
 - Calculate accuracy, precision, recall, and F1-score.
 - Generate confusion matrices and classification reports.
6. Model Comparison

- Compare the custom CNN with MobileNetV2 architecture.

8. CNN Model Design

The proposed CNN architecture consists of multiple convolutional and pooling layers followed by fully connected layers.

Architecture

- Input Layer ($224 \times 224 \times 3$)
- Conv2D (32 filters)
- MaxPooling2D
- Conv2D (64 filters)
- MaxPooling2D
- Conv2D (128 filters)
- MaxPooling2D
- Flatten Layer
- Dense Layer (128 neurons)
- Dropout Layer (0.5)
- Output Layer (3 neurons with Softmax activation)

This architecture is designed to automatically learn important visual features from cattle disease images and classify them into their respective categories.

9. Transfer Learning Model

To compare performance, MobileNetV2 architecture was selected as an additional model.

MobileNetV2 is a lightweight deep learning architecture commonly used for image classification tasks. It requires fewer parameters compared to traditional CNN models and is suitable for real-time applications.

Due to internet restrictions in the Kaggle environment, pretrained ImageNet weights were not available during experimentation. Therefore, MobileNetV2 was trained without pretrained weights and used primarily for architectural comparison.

10. Evaluation Metrics

The performance of the models is evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report

11. Expected Figures and Tables

The findings of this study will be presented through a combination of figures and tables to make the results easier to understand and interpret. Visual representations will help illustrate the learning process of the models, highlight classification performance, and provide a clear comparison between the CNN and MobileNetV2 architectures.

Expected Figures

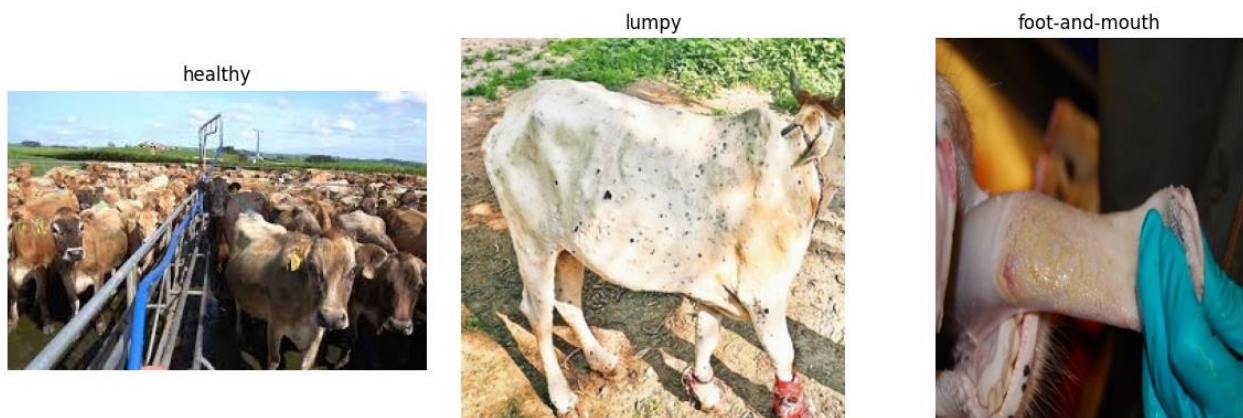


Figure 1: Sample Images from the Dataset

This figure will present example images from each category in the dataset, including Healthy, Lumpy Skin Disease, and Foot-and-Mouth Disease cattle. It will provide a visual understanding of the different classes used for model training.

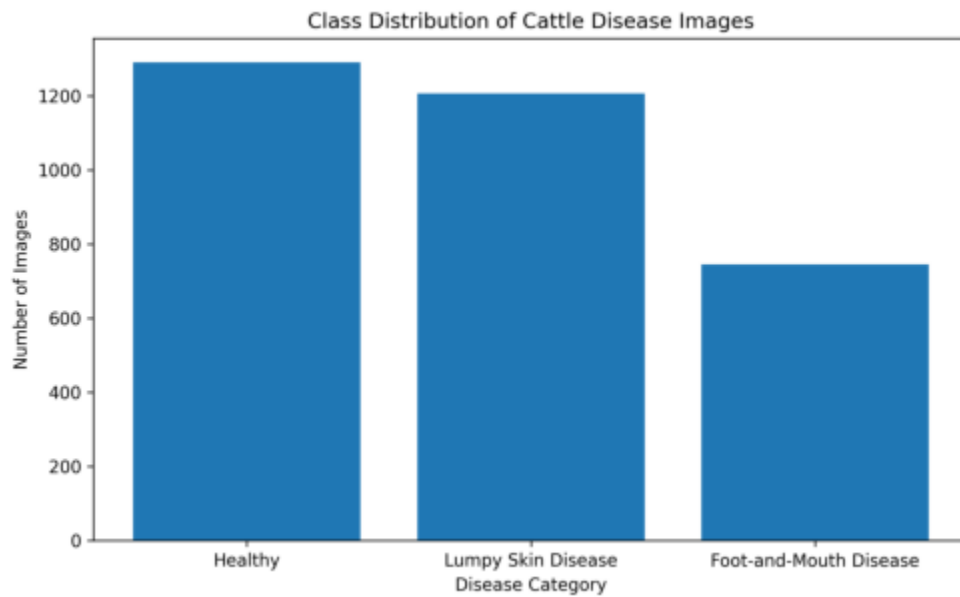


Figure 2: Class Distribution of Cattle Disease Images

This figure will show the number of images available in each class. It will help illustrate the balance of the dataset and provide an overview of the data used in the study.

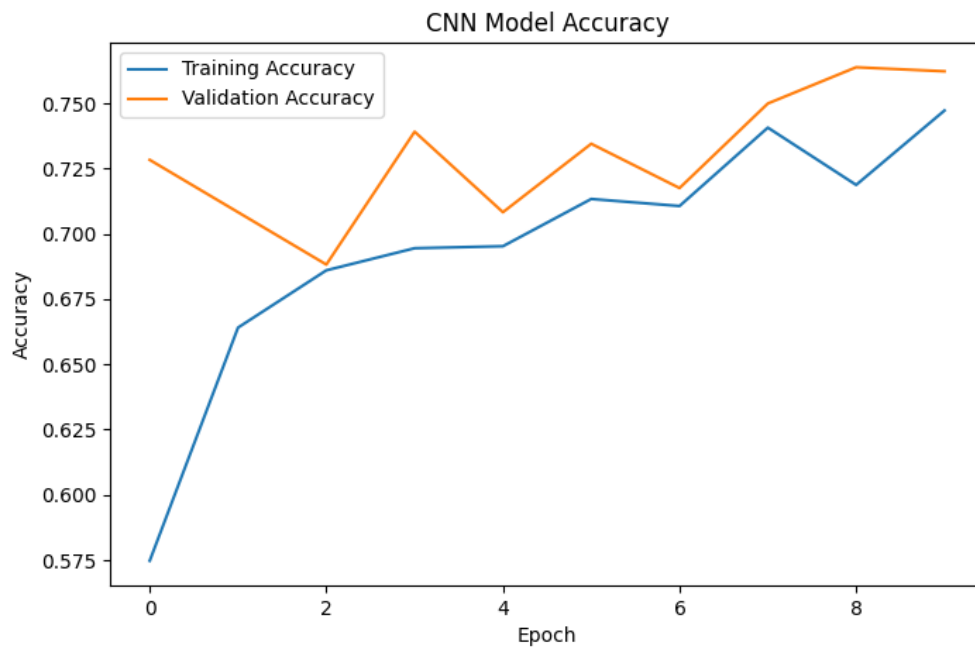


Figure 3: CNN Training Accuracy Curve

This graph will display the training and validation accuracy of the CNN model across different epochs. It will help evaluate how well the model learns from the data over time.

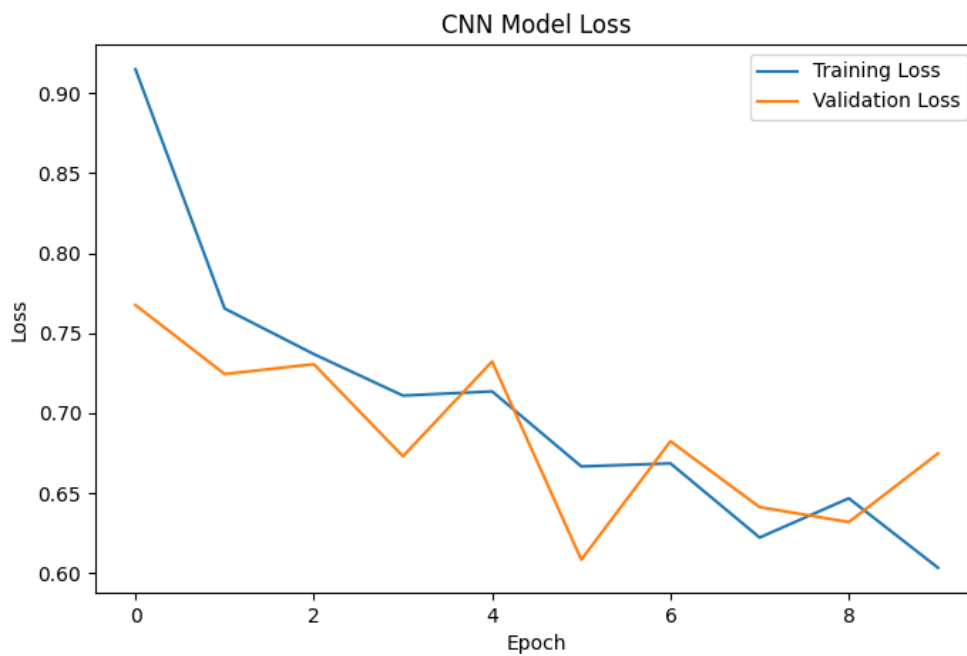


Figure 4: CNN Training Loss Curve

This figure will show the training and validation of loss values during the training process. It will be useful for assessing model convergence and identifying potential overfitting or underfitting issues.

Expected Tables

Class	Number of Images
Healthy	1,291
Lumpy Skin Disease	1,207
Foot-and-Mouth Disease	746
Total	3,244

Table 1: Dataset Class Distribution

This table will provide the number of images available for each disease category in the dataset.

Layer	Output Shape	Parameters
Conv2D (32 filters, 3×3)	(222, 222, 32)	896
MaxPooling2D	(111, 111, 32)	0
Conv2D (64 filters, 3×3)	(109, 109, 64)	18,496
MaxPooling2D	(54, 54, 64)	0
Conv2D (128 filters, 3×3)	(52, 52, 128)	73,856
MaxPooling2D	(26, 26, 128)	0
Flatten	(86,528)	0
Dense (128 neurons, ReLU)	(128)	11,075,712
Dropout (0.5)	(128)	0
Dense (3 neurons, Softmax)	(3)	387

Table 2: CNN Architecture Configuration

This table will describe the structure of the CNN model, including convolutional layers, pooling layers, dense layers, activation functions, and other important parameters.

Model Summary	Value
Total Parameters	11,169,347
Trainable Parameters	11,169,347

Non-Trainable Parameters	0
--------------------------	---

Table 3: CNN Model Summary

Table 3 summarizes the overall structure and complexity of the proposed CNN model. It presents the total number of parameters, trainable parameters, and non-trainable parameters within the network. The model contains over 11 million trainable parameters, enabling it to learn important visual features from cattle disease images and perform multi-class classification effectively.

Class	Precision	Recall	F1-Score
Foot-and-Mouth	0.79	0.77	0.78
Healthy	0.84	0.71	0.77
Lumpy	0.72	0.84	0.78

Table 4: CNN Classification Report Results

Table 4 presents the classification performance of the CNN model using evaluation metrics such as precision, recall, and F1-score for each disease category. The results indicate that the model achieved an overall classification accuracy of approximately 78%, demonstrating its ability to distinguish between healthy cattle, Lumpy Skin Disease, and Foot-and-Mouth Disease. These metrics provide a detailed assessment of the model's effectiveness across different classes.

Metric	CNN	MobileNetV2
Accuracy (%)	78.00	37.19

Table 5: Model Comparison Results

Table 5 compares the overall classification accuracy achieved by the CNN and MobileNetV2 models. The CNN model achieved an accuracy of 78.00%, outperforming MobileNetV2, which achieved 37.19%. These results suggest that the custom CNN architecture was more effective for classifying cattle diseases in the selected dataset. The comparison highlights the importance of selecting an appropriate model architecture for livestock disease detection tasks.

12. References

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. Cambridge, MA: MIT Press.

Chollet, F. (2021). *Deep Learning with Python* (2nd ed.). Manning Publications.

Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 4510–4520).

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. *Nature*, 521(7553), 436–444.

TensorFlow. (2025). *TensorFlow Documentation*. Retrieved from <https://www.tensorflow.org/>

Kaggle. (2025). *Cows Disease Dataset*. Retrieved from <https://www.kaggle.com/>